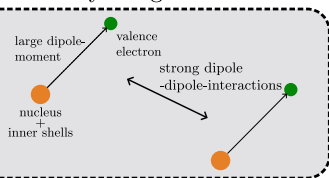
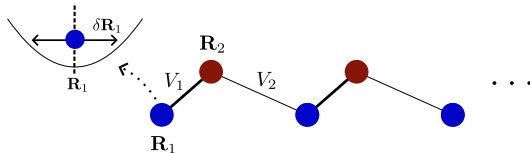


Rydberg atoms



preparing the atoms in the Rydberg states:



$$H = H_{\text{Trap}} + \sum_{i \neq j} V(\mathbf{R}_i - \mathbf{R}_j) \rightarrow H_{\text{eff}} = \sum_{i,j} U_{i,j} a_i^\dagger a_j$$

$$\text{for } i \neq j: U_{ij} = \frac{1}{m\omega} \frac{\partial^2 V(\mathbf{R}_i - \mathbf{R}_j)}{\partial \delta \mathbf{R}_i \partial \delta \mathbf{R}_j}$$

1D phonons

- $U_{i,i} = 0$
- can be mapped to the extended ssh model

2D phonons

- $U_{i,i} = 0 \leftarrow$ only if phonon directions are decoupled
- two independent nearest neighbor ssh models